揚州大學

YANGZHOU UNIVERSITY

Analysis of CMOS Inverter Circuit

Student Name	**Sifat Md Aminul Islam(西法特）**
Student ID	**238801031**
Major	**Microelectronics Science And Engineering**
Date	**2026-07-08**

1. Experiment Overview

The CMOS inverter is the most fundamental building block in digital integrated circuits and serves as the basis for designing complex CMOS logic gates. This experiment aims to analyze the electrical characteristics and switching behavior of a CMOS inverter using LTspice simulation. The performance of the inverter is evaluated by observing its voltage transfer characteristics (VTC), input-output relationship, switching threshold, and transient response. Through simulation, the behavior of complementary NMOS and PMOS transistors working together to achieve high noise immunity and low power consumption can be thoroughly examined. Understanding the operation of a CMOS inverter is essential for designing reliable and efficient digital systems.

2. Objective

The objectives of this experiment are:

- To design and simulate a CMOS inverter circuit using LTspice.
- To study the operating principle of complementary NMOS and PMOS transistors.
- To obtain the Voltage Transfer Characteristic (VTC) of the CMOS inverter.
- To analyze the switching threshold and logic inversion characteristics.
- To evaluate the power efficiency and performance of the CMOS inverter.

3. Theory

A CMOS inverter consists of one PMOS transistor connected between the supply voltage (VDD) and the output, and one NMOS transistor connected between the output and ground (GND). Both transistor gates are connected together to form the input terminal, while their drains are connected to produce the output voltage.

The operation of the CMOS inverter depends on the input voltage:

- **Input Low (Logic 0):** When the input voltage is close to 0 V, the PMOS transistor is turned ON while the NMOS transistor remains OFF. As a result, the output is connected to the supply voltage, producing a logic HIGH output.

- **Input High (Logic 1):** When the input voltage approaches VDD, the NMOS transistor turns ON while the PMOS transistor turns OFF. Consequently, the output is pulled to ground, producing a logic LOW output.

- **Transition Region:** As the input voltage increases from LOW to HIGH, both transistors conduct simultaneously for a brief interval. During this region, the output voltage rapidly changes from HIGH to LOW. The input voltage at which this transition occurs is known as the switching threshold voltage, which is an important parameter affecting noise margin and switching performance.

The Voltage Transfer Characteristic (VTC) represents the relationship between the input voltage and output voltage of the inverter. An ideal CMOS inverter exhibits a sharp transition between logic HIGH and logic LOW, providing excellent noise immunity and high voltage gain. Since only one transistor conducts during steady-state operation, CMOS inverters consume negligible static power, making CMOS technology highly suitable for low-power digital applications.

4. Software Used

- LTspice 26.0.1
- 65 nm CMOS Technology Model
- Windows Operating System

5. Methodology

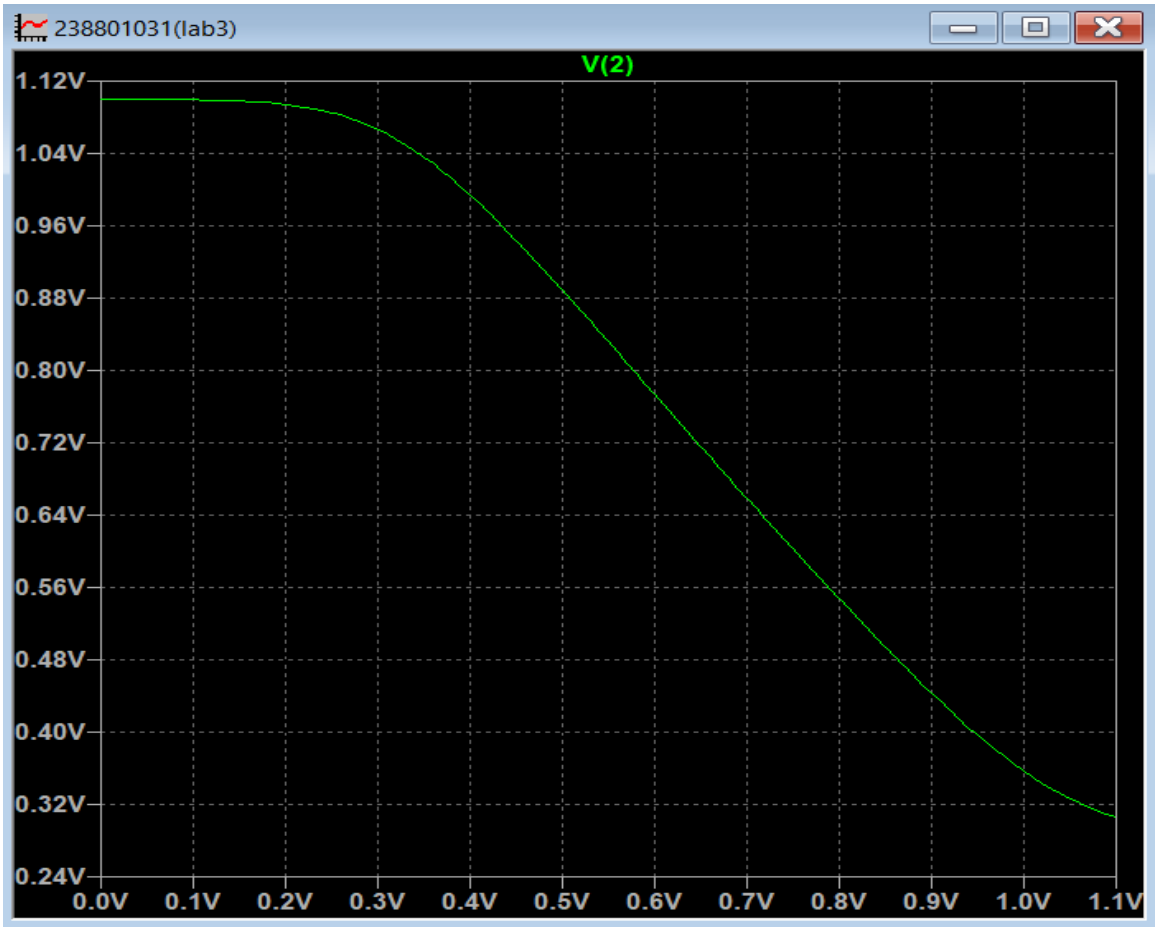
A 65 nm CMOS inverter was constructed in LTspice by connecting the PMOS source to VDD, the NMOS source to ground, and tying their gates and drains to form the input and output nodes. A DC sweep of the input voltage from 0 V to VDD was performed to plot the Voltage Transfer Characteristic (VTC) and analyze switching behavior. Additionally, a transient analysis using a pulse input verified the inversion operation and measured propagation delay, allowing the switching threshold, logic levels, and overall performance to be evaluated.

6. Programs

Output Charecteristics for E-R Inverter:

```
*output characteristics of E/R inverter for 65nm NMOS
M1 2 1 0 0 nmos w=100n l=65n m=1
R1 3 2 10k
VIN 1 0 DC 0
VDD 3 0 DC 1.1
.dc Vin 0 1.1 0.01
*model
.lib '65nm.pm'
.end
```

VTC Curve for E-R Inverter:



Output Charecteristics for E-E Inverter:

238801031(lab3)

```
* C:\Users\Sifat\Desktop\ltspice\238801031(lab3).s

* VTC for E/E Inverter

M1 out in  gnd gnd nmos W=0.1u L=0.1u
M2 out vdd vdd vdd nmos W=0.1u L=0.1u

C1 out gnd 10f

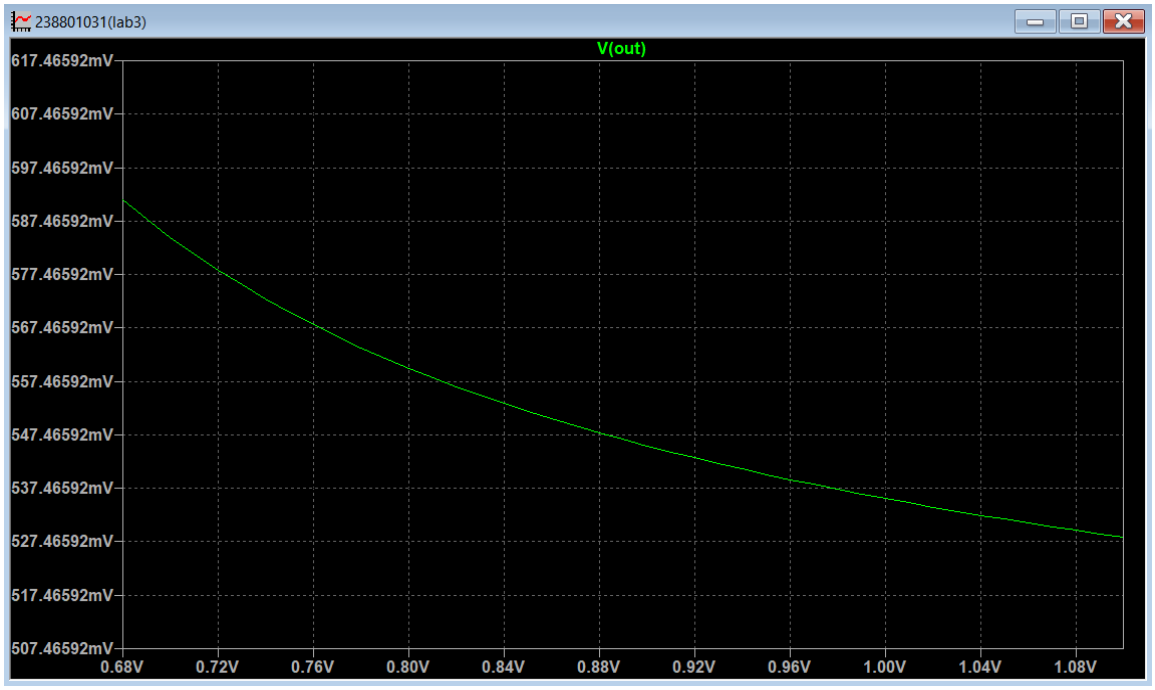
VDD vdd gnd 1.1
VIN in  gnd 0

.dc VIN 0 1.1 0.01

.inc "65nm.pm"

.end
```

VTC Curve for E-R Inverter:



VTC for Pseudo NMOS:

238801031(lab3)

```
* C:\Users\Sifat\Desktop\ltspice\238801031(lab3).sp

* VTC of Pseudo-NMOS Inverter

* NMOS Driver
M1 out in 0 0 nmos W=100n L=65n

* PMOS Load (Gate tied to GND -> Always ON)
M2 out 0 vdd vdd pmos W=100n L=65n

* Power Supply
VDD vdd 0 DC 1.1

* Input Voltage Source
VIN in 0 DC 0

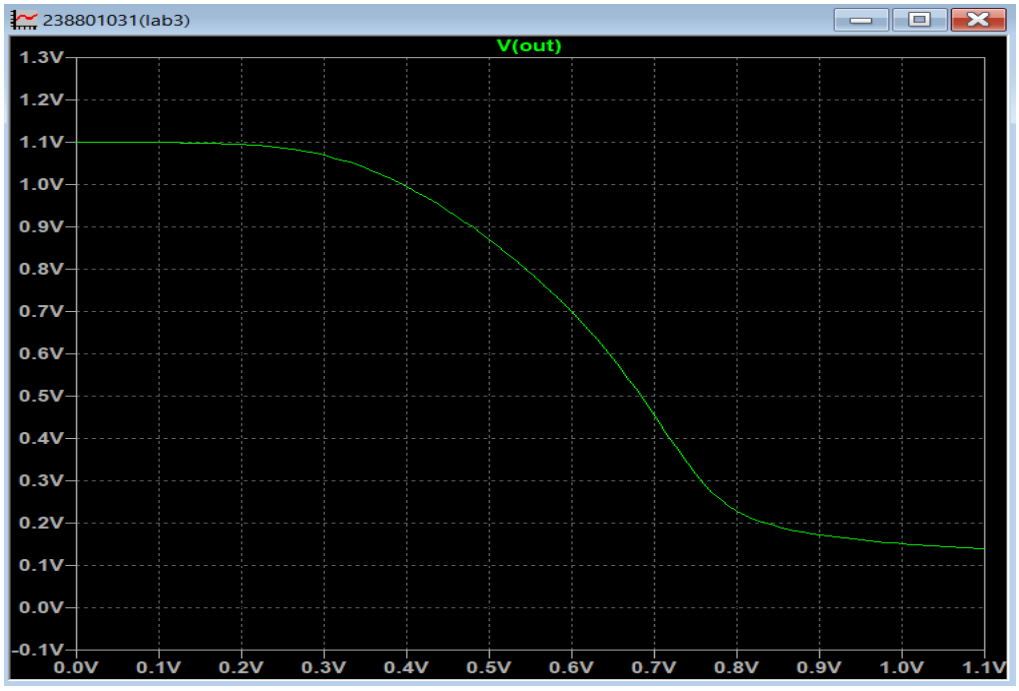
* Output Load Capacitor
CLOAD out 0 10f

* DC Sweep for VTC
.dc VIN 0 1.1 0.01

* Include 65 nm Model Library
.inc "65nm.pm"

.end
```

VTC Curve for Pseudo NMOS:



VTC for CMOS Inverter:

238801031(lab3)

```
* C:\Users\mdsor\Documents\LTspice\universal.sp
*CMOs Inverter VTC

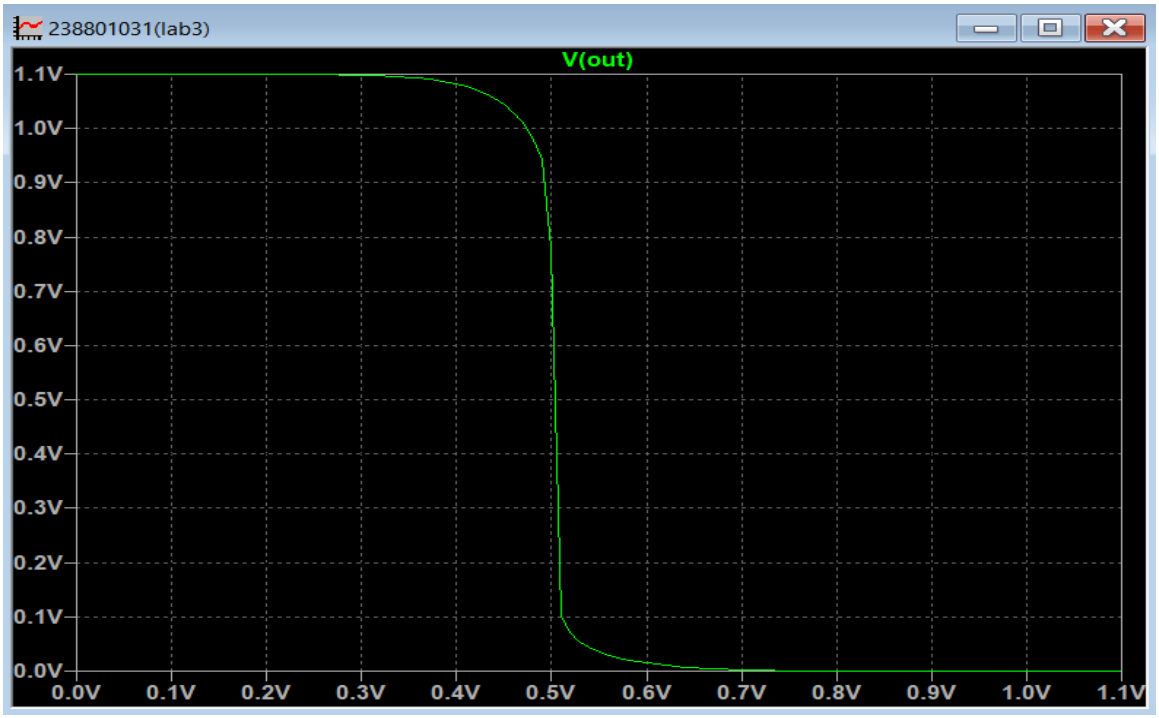
M1 out in 0 0 nmos W=0.375u L=0.25u
M2 out in vdd vdd pmos W=0.75u L=0.25u

C1 out 0 10f

VDD vdd 0 1.1
VIN in 0 0

.DC VIN 0 1.1 0.01
*model
.inc "65nm.pm"
.end
```

VTC Curve for CMOS Inverter:



7. Result:

The CMOS inverter was successfully designed and simulated using LTspice. The simulation produced the expected Voltage Transfer Characteristic and transient response, confirming the correct logic inversion operation. The switching threshold was observed within the transition region, while the output voltage changed sharply between logic HIGH and logic LOW. The experiment also verified the low static power consumption and excellent switching performance of the CMOS inverter.

8. Conclusion:

This experiment successfully analyzed the operating characteristics of a CMOS inverter through LTspice simulation. The Voltage Transfer Characteristic demonstrated efficient switching between logic states, while transient analysis verified proper inversion of the input signal. The simulation confirmed that complementary NMOS and PMOS transistors provide high noise immunity, low static power consumption, and reliable digital operation. The CMOS inverter serves as the fundamental building block of modern digital integrated circuits, and understanding its behavior is essential for the design of complex CMOS logic systems and VLSI architectures.